

TECHNOLOGY-TRANSFER PORTFOLIO — INDEX AND RANKING

4 engineering-definition packages selected for this buyer release · presented in the portfolio owner's order (the rank column preserves the disclosed portfolio ranking policy)

RANKING POLICY (disclosed)

Keys applied lexicographically: (1) KILL_TESTABILITY — a recorded numeric kill/acceptance threshold ranks above a qualitative one; (2) EVIDENCE_DEPTH — more distinct hashed external sources rank higher; (3) ENGINEERING_ARTIFACT_DEPTH — more design inputs/outputs/failure modes/verification/build steps rank higher; (4) TIME_TO_DECISIVE_EXPERIMENT — shorter recorded WP-01 effort ranks higher. Full policy in PORTFOLIO_RANKING.json. You may re-sort by your own policy from that machine-readable file.

Loop states: {'NONE': 14, 'SYNTHETIC_LOOP_VERIFIED': 1, 'REAL_LOOP_VERIFIED': 0} — Not yet demonstrated. Zero REAL events across the portfolio. The release architecture is ready to receive them (LOOP_STATE.json event queues + engine REALITY_EVENT interface).

PORTFOLIO COMPARISON TABLE (rank order)

#	Pkg	Technology	Kill test.	Ev	Art.	First decisive WP	Largest uncertainty (class)	Loop	Full record
15	P-07	Passive Drainage Priority Safety Floor	QUALIT	5	27	10 weeks	Whether floor is genuinely immune to common-cause obstruction — UNKNOWN (critica	NONE	DOWNL OAD/04 _drainag e_floor/02 dossier
14	P-24	Gravity Compensation Hydraulic Damper for Po	QUALIT	5	31	8 weeks	Comparative advantage over existing ASDs — UNKNOWN (critical for clinical adopti	SYNTH	DOWNL OAD/11 _gravity _damper /02 dossier
4	P-27-R1	Self-Referencing Piezoresistive Pressure Sen	QUANTI	5	35	16 weeks (foundry	Achievable drift rate with self-referencing — UNKNOWN (critical for clinical uti	NONE	DOWNL OAD/13 _pressur e_senso r/02 dossier
13	P-16	NIR Photovoltaic Power Delivery for Implan	QUANTI	5	28	10 weeks	Achievable power at target tissue depth (5 cm) — UNKNOWN (critical for feasibili	NONE	DOWNL OAD/08 _nir_pho tovoltaic/ 02 dossier

Ev. = distinct hashed external sources; Art. = design inputs+outputs+failure modes+verification+build steps; Kill test. QUANT = recorded numeric threshold; QUAL = qualitative. Cost range for all packages: NOT_ESTABLISHED (no quotation basis in the engineering record). Cells in this table are SUMMARIES — the authoritative record for every row is the package's engineering dossier under DOWNLOAD//.

PACKAGE ONE-LINERS (portfolio order)

04 · P-07 — Passive Drainage Priority Safety Floor (rank 15; full record: [DOWNLOAD/04_drainage_floor/02 dossier](#)). Problem: The prior generalized 30-50% obstruction-rate statement was inadequately qualified. Published studies report population- and study-specific obstruction proportions: a systematic review of 38,095 adult shunt surgeries (PubMed 37004137) found obstruction accounted for approximately 23.2% of adult shunt failures, while pediatric populations report higher proportions in some studies. Overall shunt failure rates are 40-50% at 1-2 years (PubMed 42490332), of which obstruction is one major cause but not the entirety. The exact obstruction fraction varies by patient population, study design, and follow-up duration. The dossier should cite population-specific sources rather than a universal rate. with no passive safety floor Kill: Common-cause obstruction affects both primary and floor lumens equally (no differential immunity)

08 · P-16 — NIR Photovoltaic Power Delivery for Implantable Devices (rank 13; full record: [DOWNLOAD/08_nir_photovoltaic/02 dossier](#)). Problem: Implantable sensors need power without batteries or leads Kill: Tissue attenuation at 5cm depth prevents sufficient power delivery (>1mW/cm² required at PV surface)

11 · P-24 — Gravity Compensation Hydraulic Damper for Postural Transients (rank 14; full record: [DOWNLOAD/11_gravity_damper/02 dossier](#)). Problem: Postural ICP excursions cause over-drainage and under-drainage Kill: Damper coefficient cannot be tuned to both attenuate transients AND maintain minimum drainage floor

13 · P-27-R1 — Self-Referencing Piezoresistive Pressure Sensor (rank 4; full record: [DOWNLOAD/13_pressure_sensor/02 dossier](#)). Problem: Chronic ICP monitoring sensors suffer from drift Kill: Self-referencing cannot reduce drift below 1 mmHg/month (insufficient for clinical utility)